Vigneshwararaj **Sathianarayanan**

Computer Science Engineer · Master's Student

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Profile

Master's student in Computer Science (AI & ML) at Linköping University, looking for an industry thesis in AI/ML for spring 2027. I can start in mid-January 2027 and am open to any location in Sweden. At SKF I built a predictive maintenance model that reached **92% AUC**, a data pipeline on Microsoft Fabric, and LLM tools for test data. I also fine-tuned a vision-language transformer with QLoRA and spent two years as a software engineer at NielsenIQ. I work mainly in **Python** and **PyTorch**.

Technical Skills

Machine Learning	Predictive Maintenance, Time-Series Analysis, Anomaly Detection, Classification, Clustering
Deep Learning	PyTorch, TensorFlow, PyTorch Geometric, Transformers, LSTMs, Diffusion-Based Encoders
LLMs & Generative AI	LLM Fine-Tuning (QLoRA, Custom Embeddings), Vision-Language Models, RAG, Agents
Computer Vision	CLIP Fine-Tuning, DINO Implementation, Object Detection (YOLO), MediaPipe
Other AI	Reinforcement Learning (Soft Actor-Critic), ROS 2 Nav2
Data Analysis	Statistical Analysis, Data Visualization, Sensor Data, Feature Extraction, Data Filtering
Data Platforms	Apache Spark, Delta Lake, Microsoft Fabric, Snowflake, SharePoint
Azure	AI Foundry, Azure ML, Data Lake Storage
Languages & Tools	Python, MATLAB, SQL, Java, C++, Git, Linux, JUnit

Experience

SKF Vertevo

Gothenburg, Sweden

Summer Job, Machine Learning & Data Engineering — *June 2026 - August 2026*

- **Sensor Data Analysis:** Analyzed large multi-channel sensor datasets from endurance tests on rotating machinery. Filtered and reduced high-frequency readings into per-window, per-channel features. Compared deviations across channels and found that shaft speed and power carried a stronger failure signal than bearing temperature.
- **Predictive Maintenance Model:** Built a Python model that predicts pass/fail for endurance tests: a diffusion-based time-series encoder for anomaly detection with a classifier on top. Trained it on a small set of labeled runs and reached **92% AUC**.
- **Data Pipeline:** Built a medallion pipeline on Microsoft Fabric with Spark and Delta tables. It turns structured, semi-structured and unstructured measurement data from SharePoint, Snowflake and Azure Data Lake Storage into a cleaned, structured dataset for analytics.
- **Measurement File Parsing Agent:** Built an agent on Azure AI Foundry and Azure ML that parses test-rig files whose format varies by machine and site, including Mandarin-language output. A fingerprinting step detects known formats, so the agent runs only on new formats or failed parses.
- **RAG Layer:** Designed a semantic model and RAG system over the cleaned data, used by downstream reports and a conversational data agent. It ingests PDFs, embeds their tables, images and text in one shared vector space, CLIP-style, and uses cluster-based chunking instead of fixed-size chunks.

NielsenIQ

Chennai, India

Software Engineer (Platform API) — *July 2023 - July 2025*

- **Recursive Dynamic Chars Engine:** Designed and built backend logic in the Dimensional Metadata Service and Data Factory that resolves nested product hierarchies automatically. It replaced manual SQL work and cut ticket resolution time from **2 days to real time**.
- **Partial Caching System:** Built a caching layer using payload fragmentation and hashing that cut data retrieval latency by **40%**.
- **Smart Select NLU Engine (Python, TensorFlow):** Built an engine that maps natural-language queries to complex data hierarchies, with context-aware logic for resolving ambiguous queries.
- **Testing & Quality:** Extended **JUnit** test suites for complex backend logic with other engineers, raising code coverage from **60% to 86%**.
- **Infrastructure Automation:** Co-developed a maintenance scheduler that triggers jobs from live container utilization metrics.
- **Rapid Prototyping:** Delivered working prototypes in 2-week sprints, from idea to stakeholder demo.

Projects

Fine-Tuning Qwen2.5-VL for Ancient Tamil Inscription Recognition

NLP Course Project, Linköping University — *2026*

- Built a two-stage vision-language pipeline to read extinct Tamil scripts (Tamili, Vattezhuthu) from stone inscriptions. A lightweight second model restores word boundaries.
- Fine-tuned the multimodal **transformer Qwen2.5-VL-3B** with **QLoRA** and custom embeddings on ~28,000 synthetic image-text pairs, with custom vocabulary injection for scripts absent from the base tokenizer.
- Reached **95.4 BLEU** in-distribution versus **63.0 BLEU** on simulated degraded stone-textured images.

Human-Aware Robot Navigation with Reinforcement Learning

AI Robotics Course Project, Linköping University — *2026*

- Built a custom Nav2 costmap layer and a **Soft Actor-Critic** agent that adapts personal-space parameters for a simulated mobile robot, over four reward-design cycles.
- On a small sample, collisions fell from **20% to 0%** and minimum human distance rose from 0.58 m to 0.92 m, with no loss in goal completion.

Graph Neural Networks: CGCNN in PyTorch Geometric

TDDE70 Deep Learning, Linköping University

- Implemented a message-passing layer and a full **CGCNN** (Crystal Graph Convolutional Neural Network) model in **PyTorch** Geometric, with graphs built on the fly from 3D atom positions.
- Trained it to predict atomization energy on the COLL dataset. Compared a rotation- and translation-invariant model against a non-invariant one.

Vision Foundation Models and Object Detection

Computer Vision Work

- Fine-tuned a **CLIP** model and implemented a **DINO** model.
- Used **YOLO** models for object detection.

Crowd Counting Using WiFi CSI

Bachelor Thesis, Amrita Vishwa Vidyapeetham *2023*

- Privacy-centric people counting from **WiFi Channel State Information** instead of cameras, captured with two ESP32 modules across two rooms at 2.4 and 5 GHz. Team project of four.
- Compared an ANN, an **LSTM** and a complex-valued network. The LSTM performed best, improving from **49% to 67%** accuracy with phase-rotation augmentation.
- Augmented with jittering, rotation and interpolation, and validated the augmentation with k-means and agglomerative clustering.

Webcam Gesture Recognition with MediaPipe

Deep Learning Course Project, Amrita Vishwa Vidyapeetham *2022*

- Solo project. Used MediaPipe on a live webcam feed and trained a model on gesture actions to control applications: pinch to zoom, rock-paper-scissors, and steering in a bike driving game.

Education

Linköping University *Linköping, Sweden*

M.Sc. in Computer Science (AI & ML), in progress, expected June 2027 *Aug 2025 - June 2027*

Amrita Vishwa Vidyapeetham *Coimbatore, India*

B.Tech. in Computer Science Engineering *July 2019 - June 2023*

Relevant Coursework

AI & Machine Learning Deep Learning, Neural Networks, Machine Learning, Text Mining.
Core Computer Science Software Engineering, Operating Systems, Data Structures & Algorithms, Network Science.

Languages

English Fluent (C2)

Honors & Awards

2024 **1st Prize**, Global Hackfest (NielsenIQ) - Innovation Award *Global*
2024 **2nd Prize**, Adventure Day Hackathon - Azure Monitoring
2024 **Award**, Impact Award - Rising Star (NielsenIQ)
2022 **Finalist**, Smart India Hackathon - Satellite Image Prediction *India*